

Name: **SOLUTIONS**

GCSE (1 - 9)

Parallel and
Perpendicular Lines

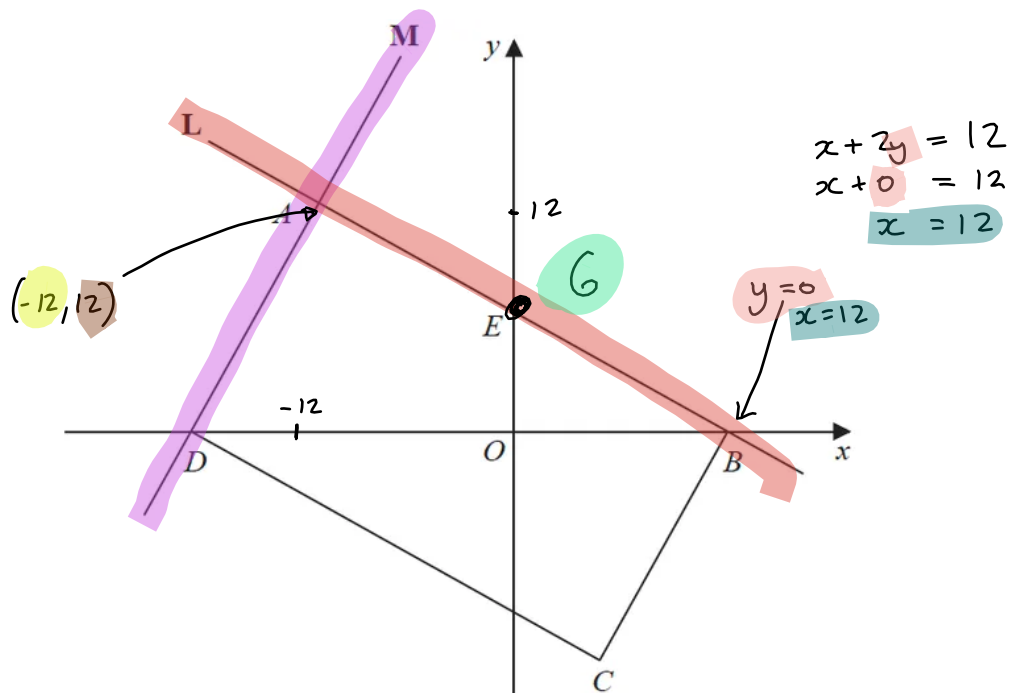
Solutions:



These questions represent all of the times that this topic has appeared in unlocked Pearson Edexcel GCSE maths papers as of July 2023. More questions can be found at [examq.co.uk](https://www.examq.co.uk)

Question	Where the question appeared
1	Edexcel, GCSE, Autumn 2017, Paper 1H, Q19
2	Edexcel, GCSE, Autumn 2017, Paper 2H, Q19
3	Edexcel, GCSE, Autumn 2019, Paper 2H, Q25
4	Edexcel, GCSE, Autumn 2020, Paper 1H, Q15
5	Edexcel, GCSE, Summer 2017, Paper 1H, Q6
6	Edexcel, GCSE, Summer 2017, Paper 1H, Q18
7	Edexcel, GCSE, Summer 2018, Paper 1H, Q19
8	Edexcel, GCSE, Summer 2019, Paper 2H, Q16
9	Edexcel, GCSE, Autumn 2021, Paper 2H, Q11

1



$ABCD$ is a rectangle.

A , E and B are points on the straight line L with equation $x + 2y = 12$

A and D are points on the straight line M .

$AE = EB$

Find an equation for M .

$$x + 2y = 12$$

$$2y = -x + 12$$

$$y = -\frac{1}{2}x + 6$$

(gradient = $-\frac{1}{2}$)
(y intercept = 6)

Line M gradient = 2 (perpendicular gradient)

$$y = 2x + c$$

$$12 = 2(-12) + c$$

$$12 = -24 + c$$

$$c = 36$$

$$y = 2x + 36$$

(Total for Question 19 is 4 marks)

[Edexcel, GCSE, Autumn 2017, Paper 1H, Q19]

Parallel and Perpendicular Lines (Questions from Edexcel GCSE Past Papers)

2

A triangle has vertices P , Q and R .

The coordinates of P are $(-3, -6)$

The coordinates of Q are $(1, 4)$

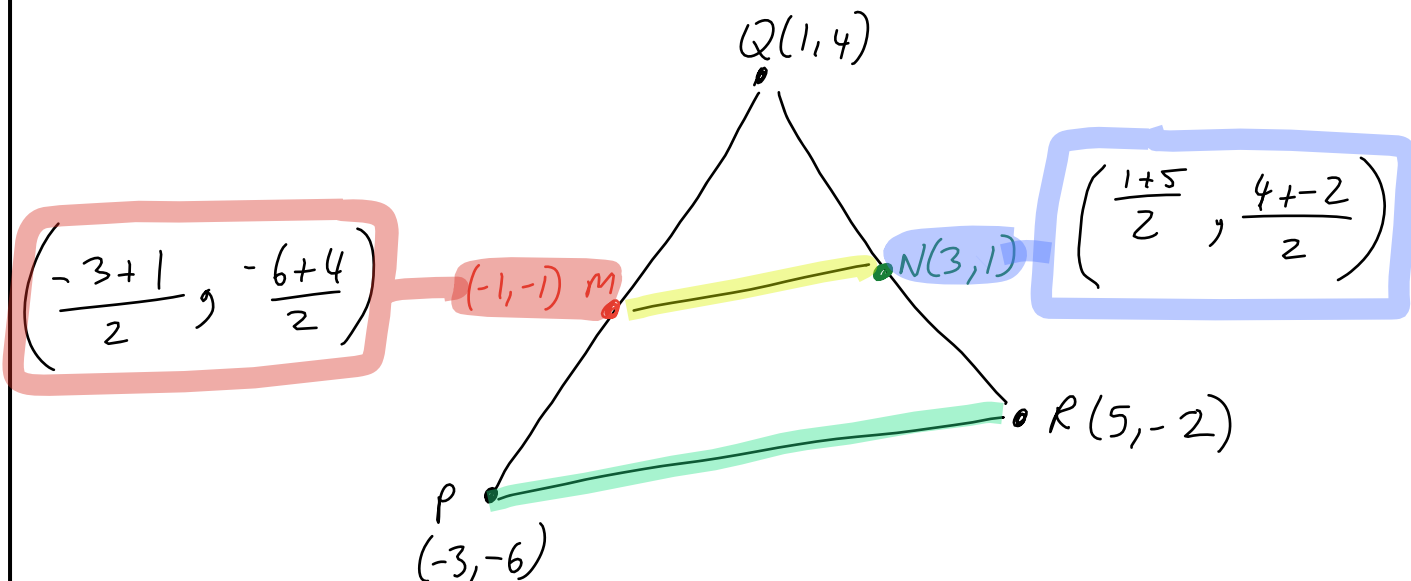
The coordinates of R are $(5, -2)$

M is the midpoint of PQ .

N is the midpoint of QR .

Prove that MN is parallel to PR .

You must show each stage of your working.



$$\text{Gradient of } PR = \frac{-2 - (-6)}{5 - (-3)} = \frac{4}{8} = \frac{1}{2}$$

$$\text{Gradient of } MN = \frac{1 - (-1)}{3 - (-1)} = \frac{2}{4} = \frac{1}{2}$$

MN and PR have the same gradient ($\frac{1}{2}$). So they are parallel.

(Total for Question 19 is 4 marks)

[Edexcel, GCSE, Autumn 2017, Paper 2H, Q19]

3

The straight line **L** has equation $3x + 2y = 17$

The point **A** has coordinates $(0, 2)$

The straight line **M** is perpendicular to **L** and passes through **A**.

Line **L** crosses the y -axis at the point **B**.

Lines **L** and **M** intersect at the point **C**.

Work out the area of triangle **ABC**.

You must show all your working.

$$3x + 2y = 17$$

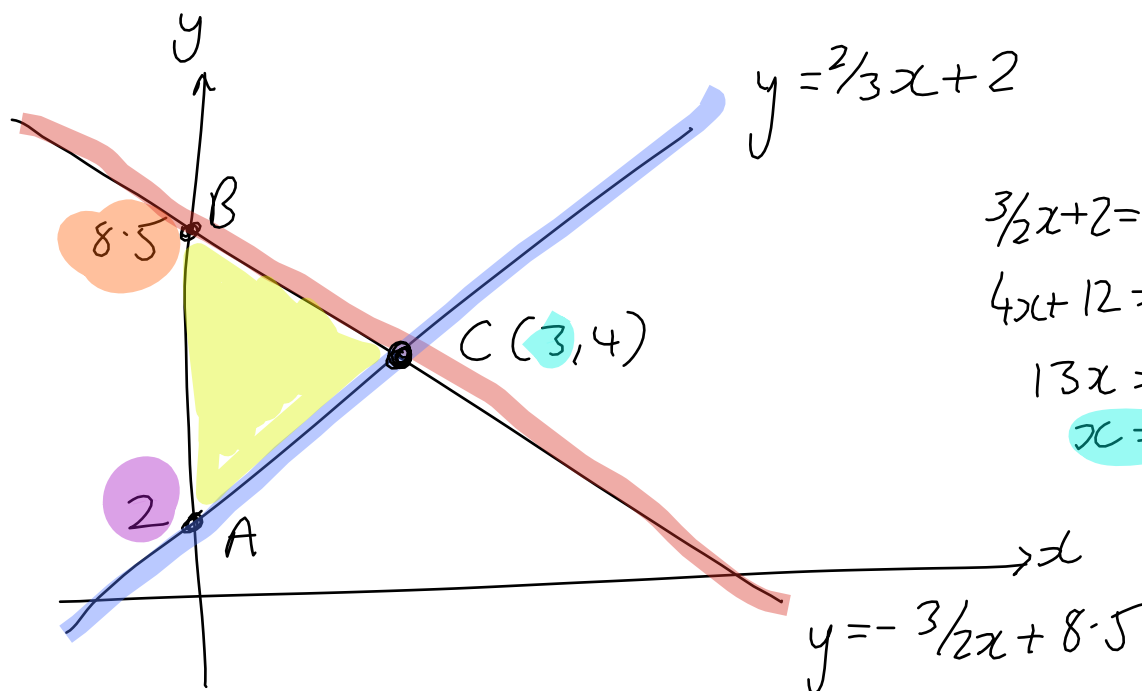
$$2y = -3x + 17$$

$$y = -\frac{3}{2}x + 8.5$$

Line **M** is perpendicular so gradient is $\frac{2}{3}$

$$y = \frac{2}{3}x + c$$

M passes through $(0, 2)$ $y = \frac{2}{3}x + 2$



$$\frac{2}{3}x + 2 = -\frac{3}{2}x + 8.5$$

$$4x + 12 = -9x + 17$$

$$13x = 5$$

$$x = \frac{5}{13}$$

$$\text{Area} = \frac{(8.5 - 2) \times 3}{2} = 9.75$$

$$9.75$$

(Total for Question 25 is 5 marks)

[Edexcel, GCSE, Autumn 2019, Paper 2H, Q25]

4

The straight line L_1 has equation $y = 3x - 4$

The straight line L_2 is perpendicular to L_1 and passes through the point (9, 5)

Find an equation of line L_2

$$L_2 \quad y = -\frac{1}{3}x + c$$

$$(5) = -\frac{1}{3}(9) + c$$

$$5 = -3 + c$$

$$c = 8$$

$$L_1 \quad m = 3$$

so

$$L_2 \quad m = -\frac{1}{3}$$

as perpendicular

$$y = -\frac{1}{3}x + 8$$

(Total for Question 15 is 3 marks)

[Edexcel, GCSE, Autumn 2020, Paper 1H, Q15]

5

The equation of the line L_1 is $y = 3x - 2$

The equation of the line L_2 is $3y - 9x + 5 = 0$

Show that these two lines are parallel.

$$L_1 \quad y = 3x - 2 \quad \text{gradient is } 3$$

$$L_2 \quad 3y - 9x + 5 = 0$$

$$3y = 9x - 5$$

$$y = 3x - \frac{5}{3}$$

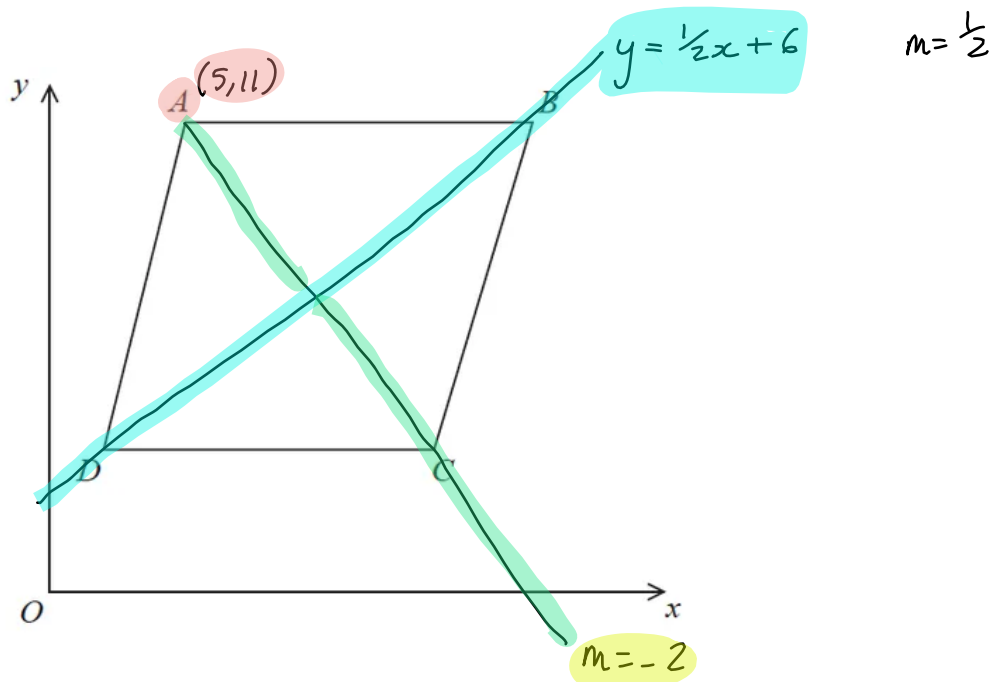
gradient is also 3

L_1 and L_2 are parallel as L_1 and L_2 have the same gradient

(Total for Question 6 is 2 marks)

[Edexcel, GCSE, Summer 2017, Paper 1H, Q6]

6



$ABCD$ is a rhombus.

The coordinates of A are $(5, 11)$.

The equation of the diagonal DB is $y = \frac{1}{2}x + 6$.

Find an equation of the diagonal AC .

AC

$$y = -2x + c$$

$(5, 11)$

$$11 = -2(5) + c$$

$$11 = -10 + c$$

$$c = 21$$

$$y = -2x + 21$$

(Total for Question 18 is 4 marks)

[Edexcel, GCSE, Summer 2017, Paper 1H, Q18]

7

The point P has coordinates $(3, 4)$

The point Q has coordinates (a, b)

A line perpendicular to PQ is given by the equation $3x + 2y = 7$

Find an expression for b in terms of a .

$$\begin{aligned} 3x + 2y &= 7 \\ 2y &= -3x + 7 \\ y &= -\frac{3}{2}x + 3.5 \end{aligned}$$

PQ

$$y = \frac{2}{3}x + c$$

$(3, 4)$

$$(4) = \frac{2}{3}(3) + c$$

$$4 = 2 + c$$

$$c = 2$$

$$y = \frac{2}{3}x + 2$$

(a, b)

$$b = \frac{2}{3}a + 2$$

$$b = \frac{2}{3}a + 2$$

(Total for Question 19 is 5 marks)

[Edexcel, GCSE, Summer 2018, Paper 1H, Q19]

8

The straight line **L** has the equation $3y = 4x + 7$

The point **A** has coordinates $(3, -5)$

Find an equation of the straight line that is perpendicular to **L** and passes through **A**.

(L)

$$3y = 4x + 7$$

$$y = \frac{4}{3}x + \frac{7}{3}$$

Perpendicular

$$y = -\frac{3}{4}x + c$$

$(3, -5)$

$$(-5) = -\frac{3}{4}(3) + c$$

$$-5 = -\frac{9}{4} + c$$

$$c = -5 + 2.25$$

$$c = -2.75$$

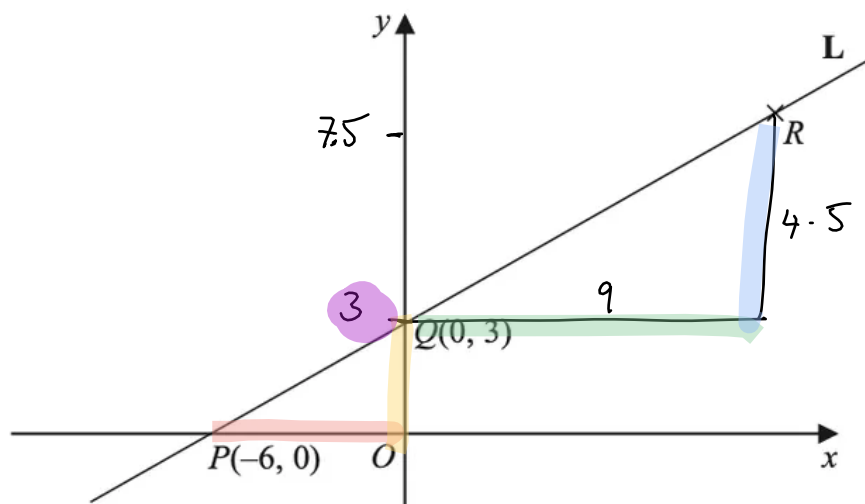
$$y = -\frac{3}{4}x - 2.75$$

(Total for Question 16 is 3 marks)

[Edexcel, GCSE, Summer 2019, Paper 2H, Q16]

9

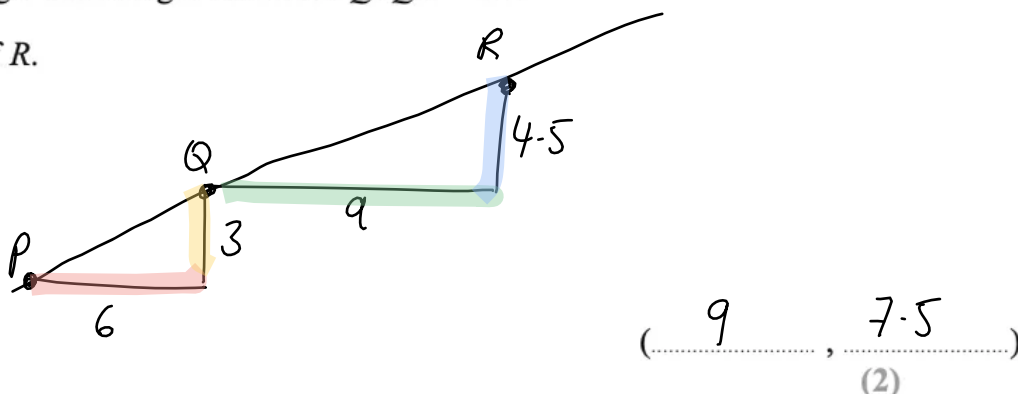
Here is a sketch of the line L.



The points $P(-6, 0)$ and $Q(0, 3)$ are points on the line L.

The point R is such that PQR is a straight line and $PQ:QR = 2:3$

(a) Find the coordinates of R.



(b) Find an equation of the line that is perpendicular to L and passes through Q.

gradient of L is $\frac{3}{6} = \frac{1}{2}$

perpendicular gradient is -2

$$y = -2x + 3$$

(Total for Question 11 is 5 marks)

[Edexcel, GCSE, Autumn 2021, Paper 2H, Q11]